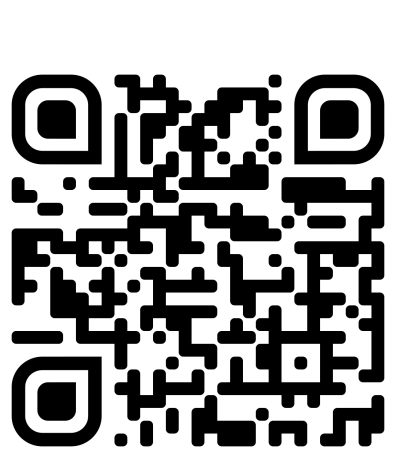




Many rays of the Submodular Cone

Georg Loho¹, Arnau Padrol², Germain Poullot³



¹ Freie Universität Berlin, Germany & University of Twente, Netherlands

² Departament de Matemàtiques i Informàtica, Universitat de Barcelona & CRM, Spain

³ Institut für Mathematik, Universität Osnabrück, Germany

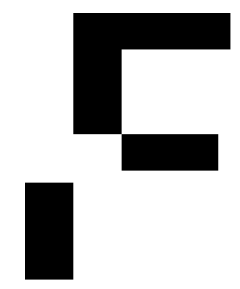
georg.loho@math.fu-berlin.de

arnau.padrol@ub.edu

germain.poullot@uni-osnabrueck.de



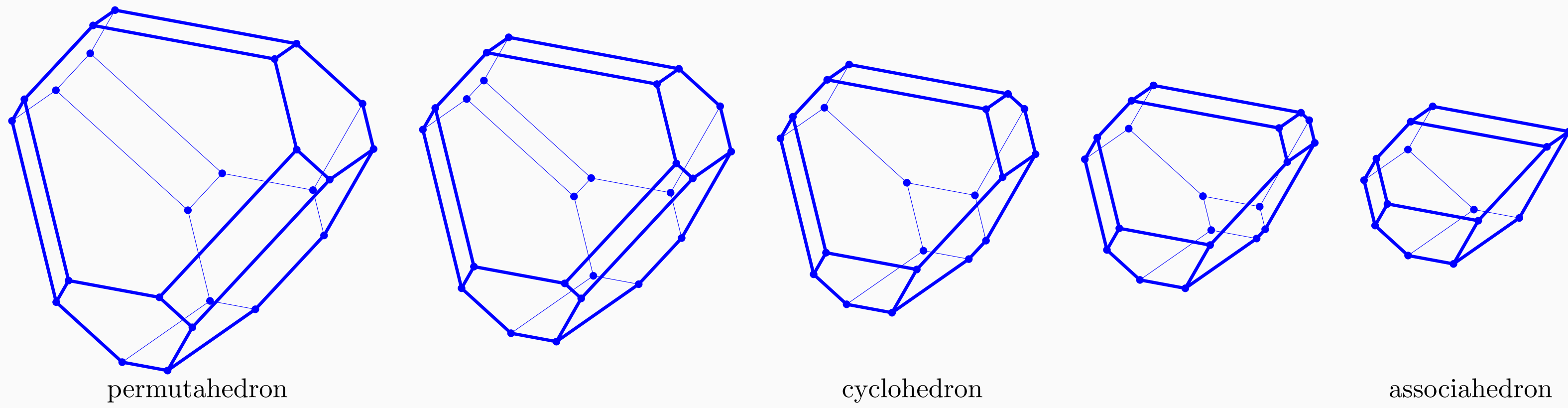
UNIVERSITAT DE
BARCELONA



FREIE
UNIVERSITÄT
BERLIN

arXiv:2510.03177

Deformation cone



Q is a *deformation* of P if $\lambda P = Q + R$ for some $\lambda > 0$ and some R

Deformation cone $\text{IDC}(P) := \{Q ; Q \text{ is deformation of } P\}$

$\text{IDC}(P)$ is closed by Minkowski sums and positive dilation: $\text{IDC}(P)$ is a cone.

Submodular cone $\mathbb{SC}_n$

Permutahedron: $\text{conv}\left\{\left(\sigma(1), \dots, \sigma(n)\right) ; \sigma \in \mathcal{S}_n\right\}$

Submodular cone: $\mathbb{SC}_n := \text{IDC}(\text{Permutohedron}_n)$
= cone submodular functions $f : 2^{[n]} \rightarrow \mathbb{R}$:

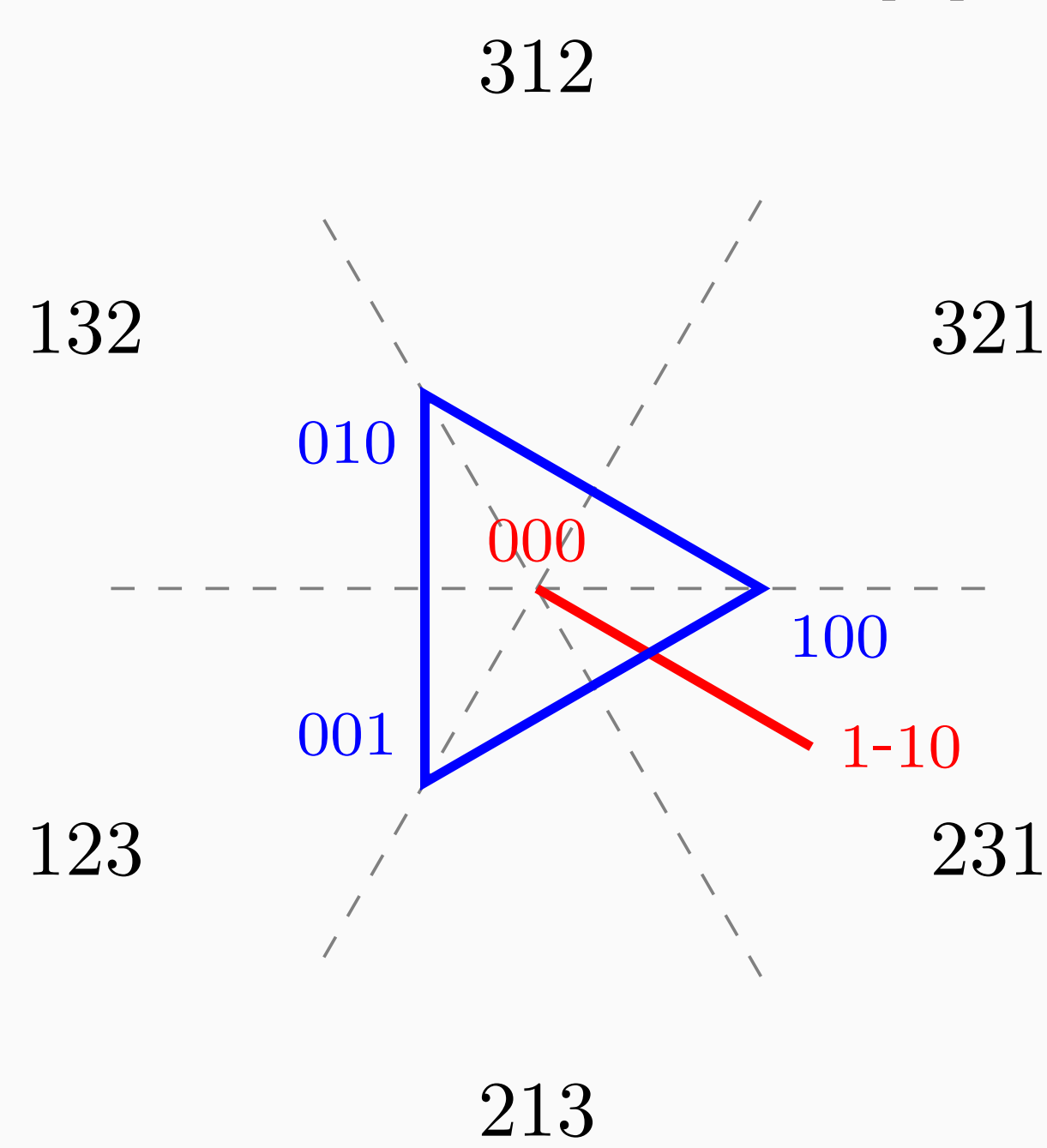
$$f(A) + f(B) \geq f(A \cap B) + f(A \cup B)$$

Edmonds '70 problem: Find (all) rays of $\mathbb{SC}_n$
a.k.a. all “indecomposable deformed permuta.”

Nguyen '86: gives $2^{2^{n-1/2} \log n + O(1)}$ rays via connected matroids (= rays on 0/1-coordinates)

GP-sum

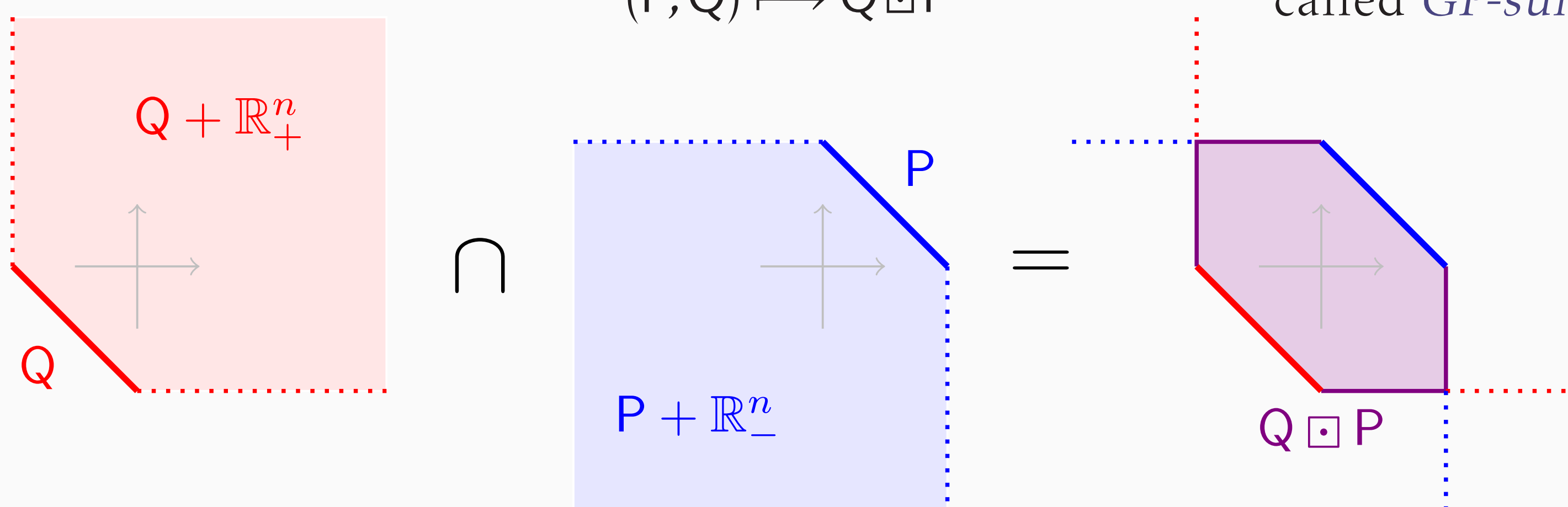
$(P, Q) \in \mathbb{SC}_n \times \mathbb{SC}_n$ is *admissible* if $\forall \sigma \in \mathcal{S}_n, i \in [n], (P^\sigma)_i \leq (Q^\sigma)_i$



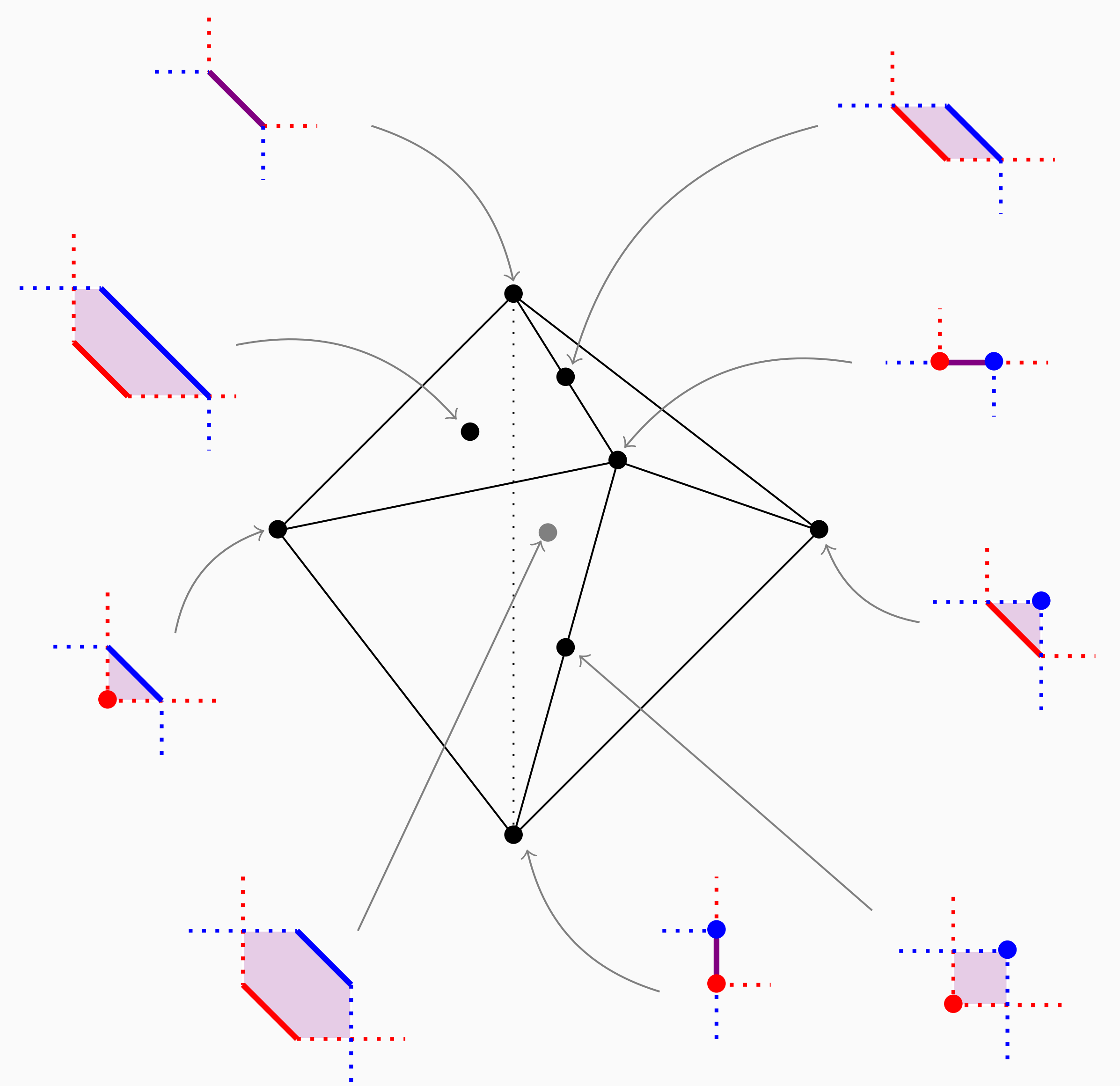
“Bijection” between $\{\text{admissible pairs} \in \mathbb{SC}_n \times \mathbb{SC}_n\}$ and $\mathbb{SC}_{n+1}$:

$$(P, Q) \mapsto Q \boxplus P$$

called *GP-sum*



$\mathbb{SC}_3 = \mathcal{C}(\text{segment}, \text{segment})$



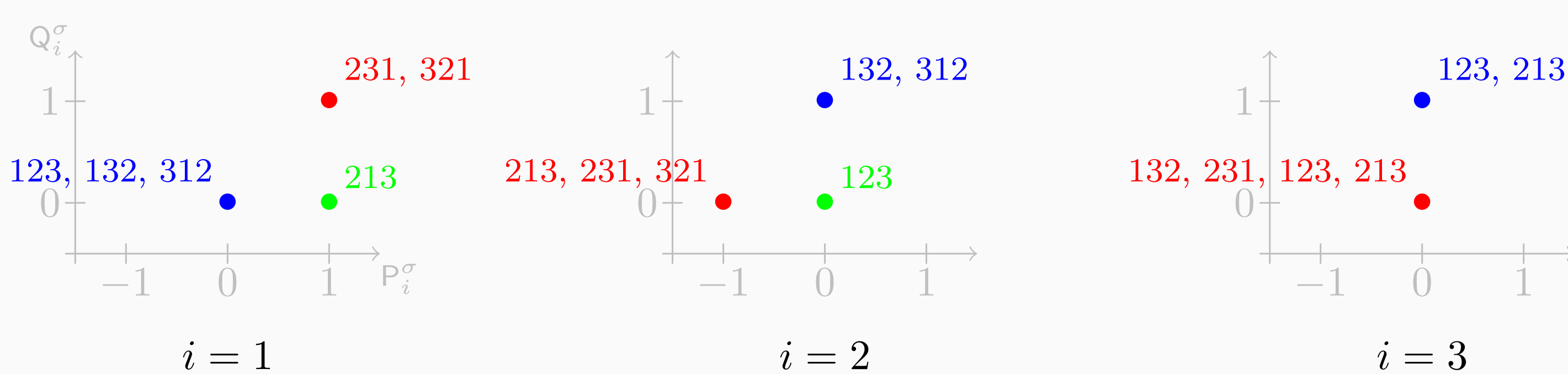
$\mathcal{C}(P, Q) = \{Q' \boxplus P' ; P', Q' \text{ deformations } P, Q \text{ and admissible pair}\}$
THM. $\mathcal{C}(P, Q)$ is a deformation cone

Seeds and fertility

Fix P, Q indecomposable, *i.e.* rays of $\mathbb{SC}_n$

Seed for (P, Q) : some λ, t such that $(\lambda Q + t) \boxplus P$ indecomposable

LEM. (P, Q) has seed iff no top-left-most point in one of the pictures:



(P, Q) is *fertile* if (P, Q) and (Q, P) have seeds

LEM. P, Q, R, S pairwise fertile $\implies (Q \boxplus P, R \boxplus S)$ fertile pair

p_n = number of pairwise fertile rays of $\mathbb{SC}_n$

THM. $p_{n+1} \geq p_n^2$

Many rays

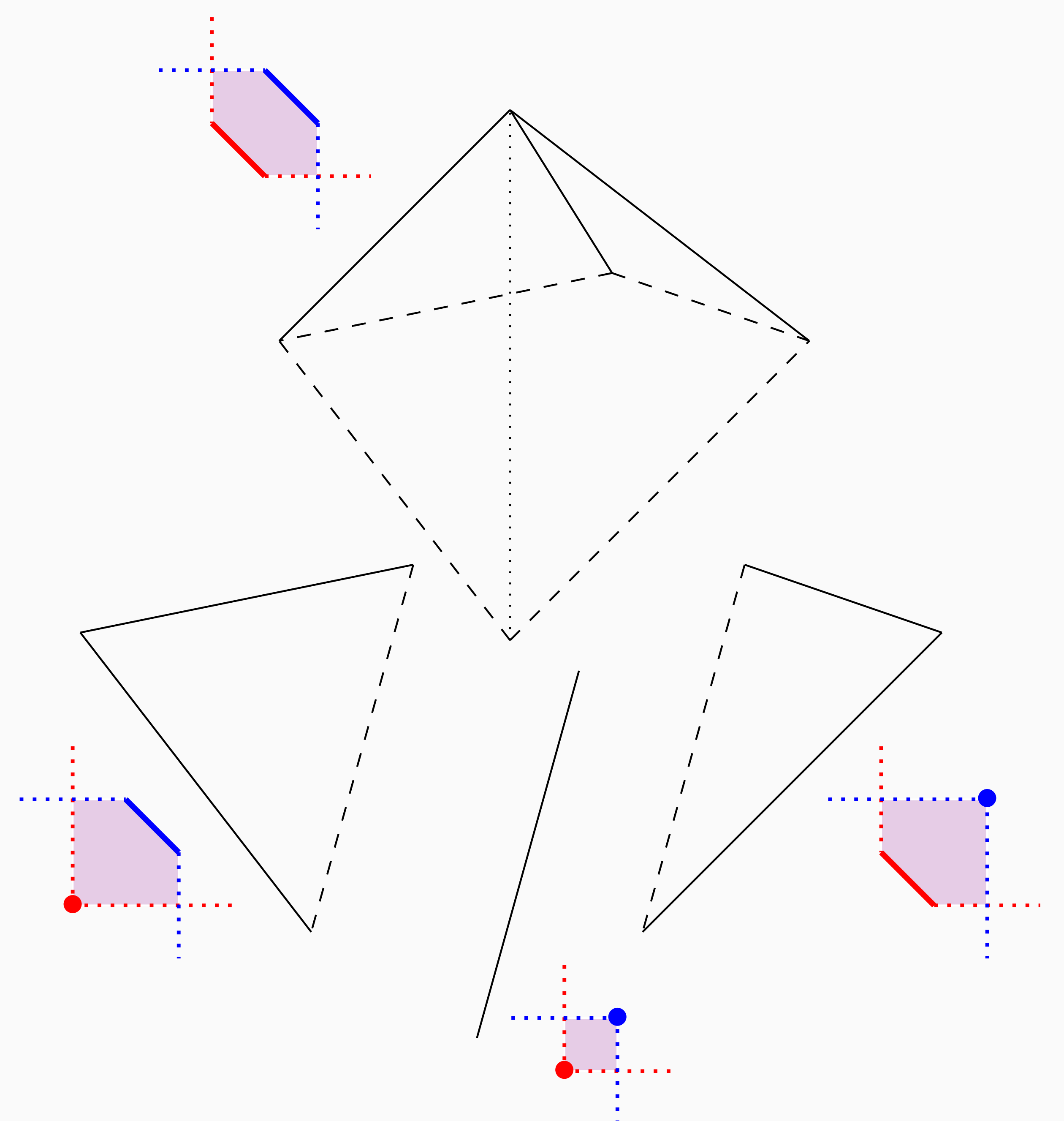
$$2^{2^{n-2}} \leq \# \text{ rays } \mathbb{SC}_n \leq n^{2^n}$$

PROOF:

rays $\mathbb{SC}_n \leq$ Upper Bound Theorem (may be improved?)

rays $\mathbb{SC}_n \geq p_n$ with $p_{n+1} \geq p_n^2$ and $p_5 \geq 656$

Going further



Our method provides partition of $\mathbb{SC}_{n+1}$ parameterized by $\mathbb{SC}_n \times \mathbb{SC}_n$.